## DELTA 3 MAX PLUS

### HTTP Connection Mode

### Set Device Functions and Get Corresponding Heartbeat

Use HTTP to set or get device function fields. The settings and requests provided in this section are open-access functions. Other non-open functions can also obtain field metrics in the same way, but cannot be used to set device functions.

PUT: /iot-open/sign/device/quota — Set device function fields (SetCmdRequest).

POST: /iot-open/sign/device/quota — Get specified heartbeat fields of the device (GetCmdRequest, GetCmdResponse).

### Examples

| Function | Set Parameter Example | Request Parameter Example | Request Response Example |
| --- | --- | --- | --- |
| Set AC output switch (true: on, false: off) | <pre>{<br>"sn": "D3M1ZA1A9H7H0136",<br>"cmdId": 17,<br>"cmdFunc": 254,<br>"dest": 2,<br>"dirDest": 1,<br>"dirSrc": 1,<br>"needAck": true,<br>"params": {<br>"cfgAcOutOpen": true<br>}<br>}</pre> | <pre>{<br>"sn": "D3M1ZA1A9H7H0136",<br>"params": {<br>"quotas": [<br>"flowInfoAcOut"<br>]<br>}<br>}</pre> | <pre>{<br>"code": "0",<br>"message": "Success",<br>"data": {<br>"flowInfoAcOut": 4<br>},<br>"eagleEyeTraceId": "",<br>"tid": ""<br>}</pre> |
| Set AC2 output switch (true: on, false: off) | <pre>{<br>"sn": "D3M1ZA1A9H7H0136",<br>"cmdId": 17,<br>"cmdFunc": 254,<br>"dest": 2,<br>"dirDest": 1,<br>"dirSrc": 1,<br>"needAck": true,<br>"params": {<br>"cfgAc2OutOpen": true<br>}<br>}</pre> | <pre>{<br>"sn": "D3M1ZA1A9H7H0136",<br>"params": {<br>"quotas": [<br>"flowInfoAc2Out"<br>]<br>}<br>}</pre> | <pre>{<br>"code": "0",<br>"message": "Success",<br>"data": {<br>"flowInfoAc2Out": 4<br>},<br>"eagleEyeTraceId": "",<br>"tid": ""<br>}</pre> |
| Set DC output switch (true: on, false: off) | <pre>{<br>"sn": "D3M1ZA1A9H7H0136",<br>"cmdId": 17,<br>"cmdFunc": 254,<br>"dest": 2,<br>"dirDest": 1,<br>"dirSrc": 1,<br>"needAck": true,<br>"params": {<br>"cfgDc12vOutOpen": true<br>}<br>}</pre> | <pre>{<br>"sn": "D3M1ZA1A9H7H0136",<br>"params": {<br>"quotas": [<br>"flowInfo12v"<br>]<br>}<br>}</pre> | <pre>{<br>"code": "0",<br>"message": "Success",<br>"data": {<br>"flowInfo12v": 14<br>},<br>"eagleEyeTraceId": "",<br>"tid": ""<br>}</pre> |
| Set energy backup switch (true: on, false: off) | <pre>{<br>"sn": "D3M1ZA1A9H7H0136",</pre> | <pre>{<br>"sn": "D3M1ZA1A9H7H0136",</pre> | <pre>{<br>"code": "0",</pre> |

| Function | Set Parameter Example | Request Parameter Example | Request Response Example |
| --- | --- | --- | --- |
| | "cmdId": 17,<br>"cmdFunc": 254,<br>"dest": 2,<br>"dirDest": 1,<br>"dirSrc": 1,<br>"needAck": true,<br>"params": {<br>"cfgEnergyBackup": {<br>"energyBackupEn": true<br>}<br>}<br>} | "params": {<br>"quotas": [<br>"energyBackupEn"<br>]<br>}<br>} | "message": "Success",<br>"data": {<br>"energyBackupEn": false<br>},<br>"eagleEyeTraceId": "",<br>"tid": ""<br>} |
| Set energy backup ratio (range: 0-50) | {<br>"sn": "D3M1ZA1A9H7H0136",<br>"cmdId": 17,<br>"cmdFunc": 254,<br>"dest": 2,<br>"dirDest": 1,<br>"dirSrc": 1,<br>"needAck": true,<br>"params": {<br>"cfgBackupReverseSoc": 41<br>}<br>} | {<br>"sn": "D3M1ZA1A9H7H0136",<br>"params": {<br>"quotas": [<br>"backupReverseSoc"<br>]<br>}<br>} | {<br>"code": "0",<br>"message": "Success",<br>"data": {<br>"backupReverseSoc": 0<br>},<br>"eagleEyeTraceId": "",<br>"tid": ""<br>} |
| Set buzzer switch (true: on, false: off) | {<br>"sn": "D3M1ZA1A9H7H0136",<br>"cmdId": 17,<br>"cmdFunc": 254,<br>"dest": 2,<br>"dirDest": 1,<br>"dirSrc": 1,<br>"needAck": true,<br>"params": {<br>"cfgBeepEn": true<br>}<br>} | {<br>"sn": "D3M1ZA1A9H7H0136",<br>"params": {<br>"quotas": [<br>"enBeep"<br>]<br>}<br>} | {<br>"code": "0",<br>"message": "Success",<br>"data": {<br>"enBeep": true<br>},<br>"eagleEyeTraceId": "",<br>"tid": ""<br>} |
| Set X-Boost switch (true: on, false: off) | {<br>"sn": "D3M1ZA1A9H7H0136",<br>"cmdId": 17,<br>"cmdFunc": 254,<br>"dest": 2,<br>"dirDest": 1,<br>"dirSrc": 1,<br>"needAck": true,<br>"params": {<br>"cfgXboostEn": true<br>}<br>} | {<br>"sn": "D3M1ZA1A9H7H0136",<br>"params": {<br>"quotas": [<br>"xboostEn"<br>]<br>}<br>} | {<br>"code": "0",<br>"message": "Success",<br>"data": {<br>"xboostEn": true<br>},<br>"eagleEyeTraceId": "",<br>"tid": ""<br>} |
| Set bypass output disable switch (true: on, false: off) | {<br>"sn": "D3M1ZA1A9H7H0136",<br>"cmdId": 17,<br>"cmdFunc": 254,<br>"dest": 2,<br>"dirDest": 1, | {<br>"sn": "D3M1ZA1A9H7H0136",<br>"params": {<br>"quotas": [<br>"bypassOutDisable"<br>] | {<br>"code": "0",<br>"message": "Success",<br>"data": {<br>"bypassOutDisable": false<br>}, |

| Function | Set Parameter Example | Request Parameter Example | Request Response Example |
| --- | --- | --- | --- |
| | <pre>"dirSrc": 1,<br>"needAck": true,<br>"params": {<br>"cfgBypassOutDisable": true<br>}<br>}</pre> | <pre>}<br>}</pre> | <pre>"eagleEyeTraceId": "",<br>"tid": ""<br>}</pre> |
| Set charge upper limit (range: 50-100) | <pre>{<br>"sn": "D3M1ZA1A9H7H0136",<br>"cmdId": 17,<br>"cmdFunc": 254,<br>"dest": 2,<br>"dirDest": 1,<br>"dirSrc": 1,<br>"needAck": true,<br>"params": {<br>"cfgMaxChgSoc": 80<br>}<br>}</pre> | <pre>{<br>"sn": "D3M1ZA1A9H7H0136",<br>"params": {<br>"quotas": [<br>"cmsMaxChgSoc"<br>]<br>}<br>}</pre> | <pre>{<br>"code": "0",<br>"message": "Success",<br>"data": {<br>"cmsMaxChgSoc": 90<br>},<br>"eagleEyeTraceId": "",<br>"tid": ""<br>}</pre> |
| Set discharge lower limit (range: 0-30) | <pre>{<br>"sn": "D3M1ZA1A9H7H0136",<br>"cmdId": 17,<br>"cmdFunc": 254,<br>"dest": 2,<br>"dirDest": 1,<br>"dirSrc": 1,<br>"needAck": false,<br>"params": {<br>"cfgMinDsgSoc": 30<br>}<br>}</pre> | <pre>{<br>"sn": "D3M1ZA1A9H7H0136",<br>"params": {<br>"quotas": [<br>"cmsMinDsgSoc"<br>]<br>}<br>}</pre> | <pre>{<br>"code": "0",<br>"message": "Success",<br>"data": {<br>"cmsMinDsgSoc": 20<br>},<br>"eagleEyeTraceId": "",<br>"tid": ""<br>}</pre> |

### Complete Heartbeat Response

| Field | Type | Description |
| --- | --- | --- |
| pow_get_ac_in | float | Real-time power of AC input port (W) |
| pow_get_pv | float | Real-time power of PV/car input (W) |
| pow_get_pv2 | float | Real-time power of PV/car 2 input (W) |
| powGetAcOutList.powGetAcOutItem | float | Real-time power of each AC output port (W) |
| pow_get_12v | float | Real-time power of 12V DC output port (W) |
| pow_get_12v_list.pow_get_12v_item | <pre>[float]</pre> | Real-time output power of Anderson port (W) |
| pow_get_typec1 | float | Real-time power of USB-C output port USB-C1 (W) |
| pow_get_typec2 | float | Real-time power of USB-C output port USB-C2 (W) |
| pow_get_typec3 | float | Real-time power of USB-C output port USB-C3 (W) |
| pow_get_qcusb1 | float | Real-time power of USB-A output port USB-1 (W) |
| pow_get_qcusb2 | float | Real-time power of USB-A output port USB-2 (W) |

| Field | Type | Description |
| --- | --- | --- |
| flow_info_ac_out | uint | AC output switch (on when value is not 4) |
| flow_info_ac2_out | uint | AC2 output switch (on when value is not 4) |
| flow_info_12v | uint | DC output switch (on when value is not 4) |
| pow_in_sum_w | float | Real-time input power of the device (W) |
| pow_out_sum_w | float | Real-time output power of the device (W) |
| cms_batt_soc | float | Current battery level (%) |
| cms_chg_dsg_state | uint | Current charging/discharging/idle status (0-idle, 1-discharging, 2-charging) |
| cms_dsg_rem_time | uint | Remaining usable time at current battery level (min) |
| cms_chg_rem_time | uint | Time required to fully charge the battery (min) |
| energy_backup_en | bool | Energy backup switch status (0-off, 1-on) |
| backup_reverse_soc | uint | Backup ratio setting (%) |
| en_beep | bool | Buzzer switch status (0-off, 1-on) |
| xboost_en | bool | X-Boost switch status (0-off, 1-on) |
| bypass_out_disable | bool | Bypass output disable switch status (0-not disabled by default, 1-disabled) |
| cms_max_chg_soc | uint | Charge upper limit setting range (%) |
| cms_min_dsg_soc | uint | Discharge lower limit setting range (%) |

**Example**


```
{
    "powGetPv": 102,
    "cmsChgRemTime": 793,
    "energyBackupEn": false,
    "powGetAcIn": 0,
    "flowInfoAcOut": 4,
    "bypassOutDisable": true,
    "powGet12v": 0,
    "backupReverseSoc": 0,
    "cmsBattSoc": 49,
    "powOutSumW": 0,
    "powGetAcOutList.powGetAcOutItem": [
        0,
        0,
        0,
        0,
        0
    ],
    "xboostEn": false,
    "cmsMaxChgSoc": 76,
    "cmsMinDsgSoc": 24,
    "flowInfo12v": 4,
    "powGetPv2": 0,
    "cmsChgDsgState": 2,
    "flowInfoAc2Out": 4,
    "enBeep": true,
    "powGetQcusb1": 0,
    "powGetQcusb2": 0,
    "powGet12vList.powGet12vItem": [
        0,
        0
    ],
    "powGetTypec3": 0,
    "powInSumW": 102,
    "powGetTypec1": 0,
    "powGetTypec2": 0,
```

```
    "cmsDsgRemTime": 12927
}
```


### MQTT Connection Mode

### Command Set Topic & Response Topic

| Purpose | Topic | From | To |
| --- | --- | --- | --- |
| Set device function | /open/${certificateAccount}/${sn}/set | app | device |
| Device responds with set result | /open/${certificateAccount}/${sn}/set_reply | device | app |

### Example

#### Set command format


```
{
    "sn": "D3M1ZA1A9H7H0136",
    "id": 1715152800124,
    "version": "1.0",
    "cmdId": 17,
    "cmdFunc": 254,
    "dest": 2,
    "dirDest": 1,
    "dirSoc": 1,
    "needAck": true,
    "params": {
        "cfgBeepEn": true
    }
}
```


### Heartbeat Report

| Purpose | Topic | From | To |
| --- | --- | --- | --- |
| Device heartbeat report | /open/${certificateAccount}/${sn}/quota | device | app |

### Example


```
{
    "powGet12vList": {
        "powGet12vItem": [
            0,
            0
        ]
    },
    "powGetPv": 102,
    "cmsChgRemTime": 793,
    "energyBackupEn": false,
    "powGetAcIn": 0,
    "powGetAcOutList": {
        "powGetAcOutItem": [
            0,
            0,
            0,
            0,
            0
        ]
    },
    "flowInfoAcOut": 4,
    "bypassOutDisable": true,
    "powGet12v": 0,
    "cmsBattSoc": 49,
```

```
    "backupReverseSoc": 0,
    "powOutSumW": 0,
    "xboostEn": false,
    "cmsMaxChgSoc": 76,
    "cmsMinDsgSoc": 24,
    "flowInfo12v": 4,
    "powGetPv2": 0,
    "cmsChgDsgState": 2,
    "flowInfoAc2Out": 4,
    "enBeep": true,
    "powGetQcusb1": 0,
    "powGetQcusb2": 0,
    "powGetTypec3": 0,
    "powInSumW": 102,
    "powGetTypec1": 0,
    "powGetTypec2": 0,
    "cmsDsgRemTime": 12927
}
```


### Device Status

| Purpose | Topic | From | To |
| --- | --- | --- | --- |
| Device status report | /open/${certificateAccount}/${sn}/status | device | app |

### Example


```
{
    "id": "945718406",
    "params": {
        "status": 0
    },
    "timestamp": 1778308183516,
    "version": "1.0"
}
```


## Was this page helpful?